The University of Texas at Austin
Mechanical Engineering
Cockrell School of Engineering

ME397: Algorithms for Sensor-Based Robotics (ASBR)
Prof. Farshid Alambeigi, Spring 2026

Homework/Programming Assignment #2 **Homework/midterm Due: 03/24/2026- 3:30 PM**

Name/EID: Noah Manasterli / njm3522
Email: njm3522@my.utexas.edu
Signature (*required*) Noah Manasterli
I/We have followed the rules in completing this Assignment.

Name/EID: Jamie Moseley / jbm4577
Email: jbm4577@my.utexas.edu
Signature (*required*) James Moseley
I/We have followed the rules in completing this Assignment.

Question	Points	Total
HA 1	**25**	
HA 2	**25**	
HA 3	**25**	
HA 4	**25**	
PA	**100**	
PA. k (Bonus)	**15**	
PA. m (Bonus)	**30**	
Presentation* (Bonus)	**20**	

Instruction:

1. Remember that this is a graded assignment. It is the equivalent of a **midterm take-home exam**.
2. *** You should present the results of the PA in the class** and receive extra bonus depending on the quality of your presentation!
3. **For PA questions**, you need to write a report showing how you derived your equations, describes your approach, test functions, and discusses the results. You should show your test results for each function.
4. You are to work **alone** or **in teams of two** and are **not to discuss the problems with anyone** other than the TAs or the instructor.
5. **You may use large language models (LLMs) such as ChatGPT or Gemini. If you do so, you must clearly state: *"I used ChatGPT to assist in responding to this assignment."* In addition, you must include the prompts you used and briefly describe how you validated the model's responses. The full sequence of prompts should be appended to your submission.**

6. It is open book, notes, and web. But you should cite any references you consult.
7. Unless I say otherwise in class, it is due before the start of class on the due date mentioned in the Assignment.
8. **Sign and append** this score sheet as the first sheet of your assignment.
9. Remember to submit your assignment in Canvas.

THA2 Programming Assignment

Functions, Tests, and Results

All functions were validated using the automated test suite in `test_THA2.m`. The robot used throughout is the **KUKA KR120 R2500 Pro (Quantec Nano)**, with kinematic parameters derived from the manufacturer-supplied URDF (`kr120r2500pro.urdf`, `kuka_experimental` ROS package). **All 64 automated tests pass.**

Part (a) — Robot Kinematic Parameters

Part (a) involves establishing the complete kinematic model of the KUKA KR120 R2500 Pro in the Product of Exponentials (PoE) framework. There is no standalone algorithm function for this part—the deliverable is the parameter struct produced by `KR120_params.m`, which all subsequent functions consume.

Approach

All parameters are derived directly from `kr120r2500pro.urdf` (`kuka_experimental` ROS package). The six link dimensions are read from the URDF `joint origin` fields:

Parameter	Value	Source
d_1	0.675 m	joint_a1 z
a_1	0.350 m	joint_a2 x
a_2	1.150 m	joint_a3 x
a_3	1.000 m	joint_a4 x
δz	−0.041 m	joint_a4 z (wrist z-drop)
d_6	0.215 m	tool0 x

The wrist centre is computed as wc = $[a_1+a_2+a_3;\ 0;\ d_1+\delta z] = [2.500;\ 0;\ 0.634]$ m. Joints 4, 5, and 6 are co-located at this point, forming a spherical wrist.

The home configuration M is the end-effector pose at $\theta = 0$. The `tool0` fixed joint in the URDF specifies `rpy="0 π/2 0"`, so M carries a $R_Y(+90°)$ rotation and end-effector position [2.715; 0; 0.634]:

$$M = \begin{bmatrix} 0 & 0 & 1 & 2.715 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0.634 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Each space screw axis is built from its joint axis direction ω_i and a point q_i on the axis using $v_i = -\omega_i \times q_i$. Joint axes are taken directly from the URDF: joints 1, 4, and 6 rotate about negative-axis directions (−Z, −X, −X), which is important to match the physical robot's behaviour. The body screw axes B□e□□ are derived automatically via $B_i = \mathrm{Ad}(M^{-1})\cdot S_i$, so they never need to be specified manually.

Validation

The parameter file is implicitly validated by every subsequent test: if M, S□^{e}□□, or B□^{e}□□ were incorrect, the FK identity test (b1) and the space/body agreement tests (c2–c3) would immediately fail. All 64 tests pass, confirming the parameters are internally consistent and geometrically correct.

Part (b) — FK_space.m

Approach

The space-form forward kinematics computes the end-effector transform as:

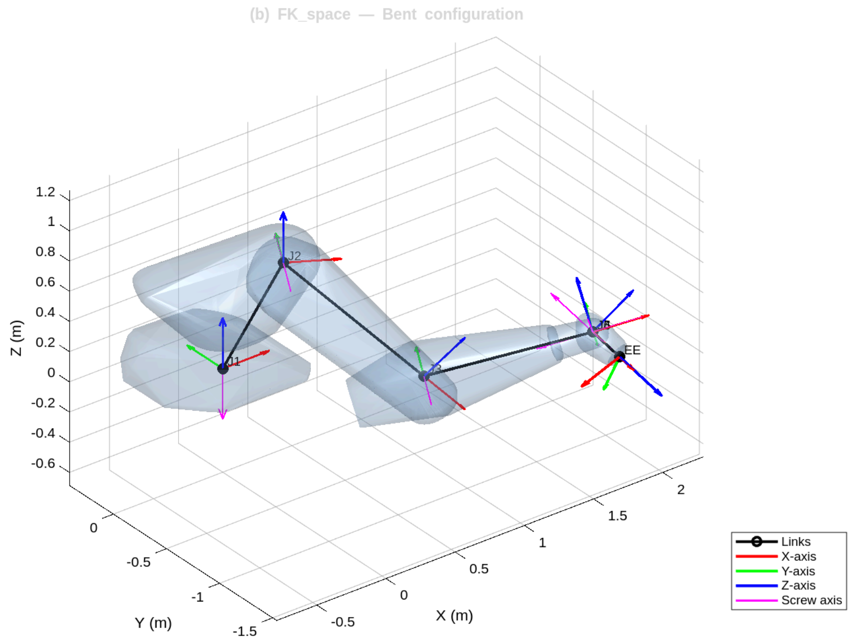
$$T(\theta) = e\^([S_1]\theta_1) \cdots e\^([S_6]\theta_6)\ M$$

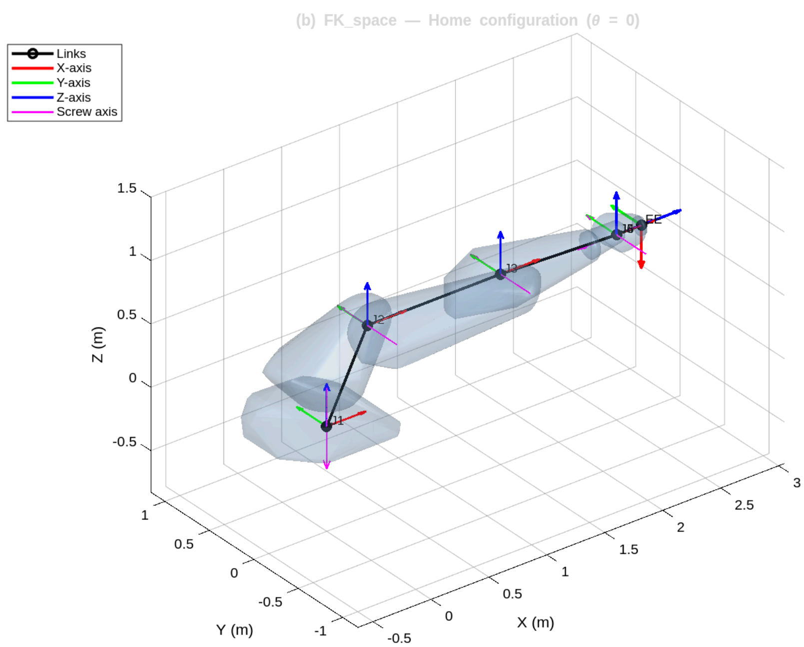
using the matrix exponential `MatrixExp6` applied to each successive screw axis. Partial transforms $T^{fr}{}_{,i}$ are accumulated to recover joint-frame positions for visualization.

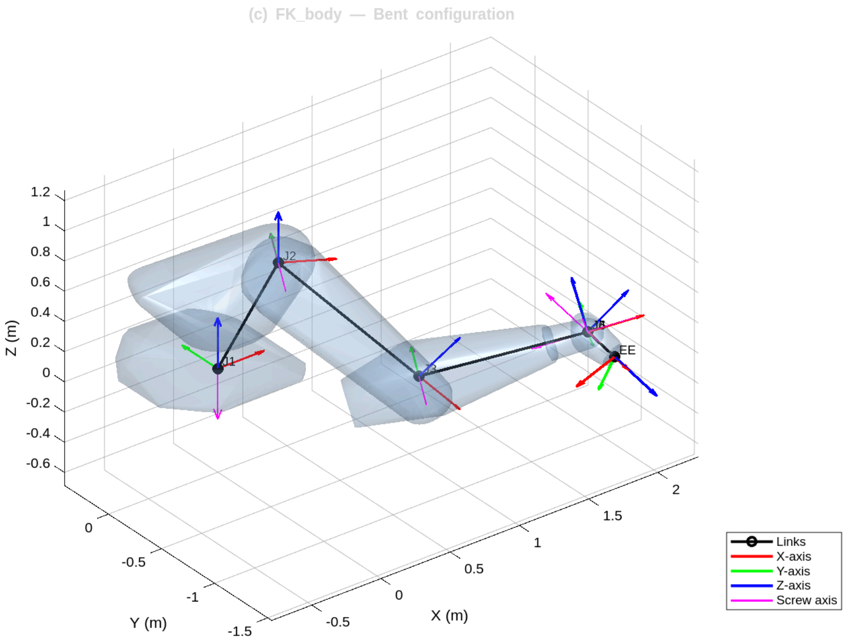
Tests and Results

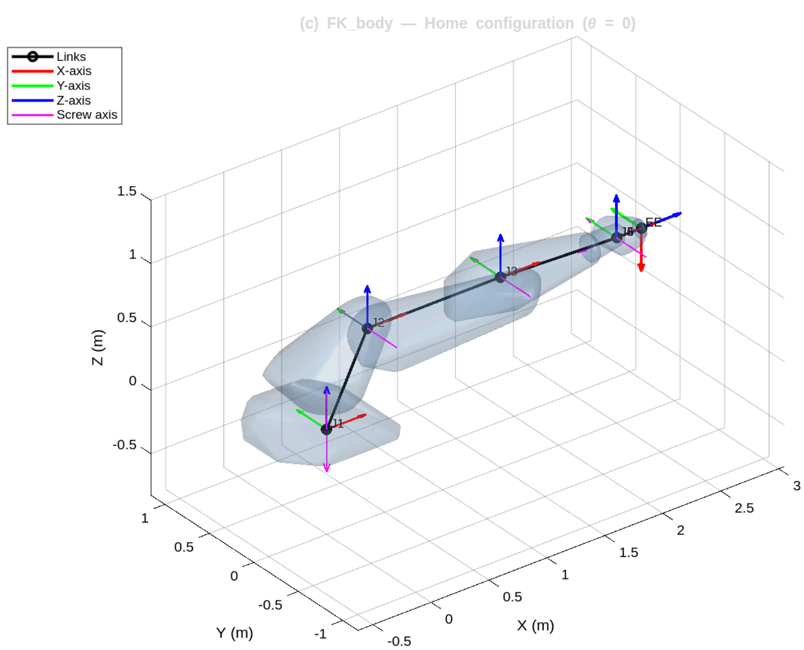
Test	Condition	Result
b1	$\|T(0) - M\|_x$	0 (exact)
b2	$\|R(0) - M_r{}^{ot}\|_x$	0 (exact)
b3	J1 rotation π/2: XY radius preserved	Δ = 0 m
b4	J1 rotation π/2: Z height preserved	Δ = 0 m
b5	J2 rotation π/2: EE position = [0.309, 0, −1.690] m	error = 3.1×10^{-16} m

At θ = 0, `FK_space` returns exactly M, confirming the screw axes and M are consistent. Test b2 verifies the home orientation is R_γ(+90°), matching the `tool0` fixed joint. Test b5 analytically derives the expected end-effector position after a pure joint-2 sweep—accounting for the 41 mm z-drop—and confirms the result to numerical precision ($\varepsilon \approx 3.1 \times 10^{-16}$ m). Visualization was verified interactively, showing joint frames and screw axes overlaid on the robot mesh.


(b) FK_space — Bent configuration
Z (m)
Y (m)
X (m)
J1
J2
J3
EE
Links
X-axis
Y-axis
Z-axis
Screw axis


(b) FK_space — Home configuration (θ = 0)
Links
X-axis
Y-axis
Z-axis
Screw axis
Z (m)
Y (m)
X (m)
J1
J2
J3
EE





Part (c) — FK_body.m

Approach

The body-form FK computes the end-effector transform as:

$$T(\theta) = M\, e^{\wedge}([B_1]\theta_1) \cdots e^{\wedge}([B_6]\theta_6)$$

where body screw axes $B_i = Ad(M^{-1})S_i$ are pre-computed in `KR120_params.m`.

Tests and Results

Test	Condition	Result
c1	$\lVert T^{body}(0) - M \rVert_x$	0 (exact)
c2	$\lVert T_{□□aoe} - T^{body} \rVert_x$, arbitrary θ	8.95×10^{-16}
c3	Max error over 20 random configs	1.15×10^{-15}
c4	$\lVert R^T R - I \rVert_x$	1.26×10^{-16}
c5	det(R) − 1	0 (exact)
c6	Bottom row error	0 (exact)

The body and space forms agree to within floating-point precision ($\sim 10^{-15}$) across all 20 randomly sampled configurations, confirming both implementations are correct. The output is verified to be a valid element of SE(3): $R^T R = I$, $\det(R) = 1$, and the homogeneous row is [0 0 0 1].

Part (d) — Space Jacobian (J_space)

The function `J_space(Slist, thetalist)` computes the 6×n space Jacobian, where column i is the i-th space screw axis transformed into the current frame of all preceding joints:

$$J_{□}^{-1} = Ad(e\hat{}([S_1]\theta_1) \cdots e\hat{}([S_{i-1}]\theta_{i-1})) \cdot S_i$$

Column 1 is always S_1 unchanged (no preceding joints). Each subsequent column is the original screw axis S_i “moved” by the cumulative transform of all joints before it.

Tests and Results

See part (e)

Part (e) — J_space.m and J_body.m

Approach

The space Jacobian is built column-by-column. The i-th column is given by:

$$J_{□}^{-1} = Ad(T_1 \cdots T_{i-1})\ S_i$$

The body Jacobian follows from the adjoint relationship: $J^b = Ad(T^{-1})\ J_{□}$.

Tests and Results

Test	Condition	Result
d1	Column 1 of $J_{□}(0)$ equals S_1	error = 0
d2	$\lVert J_{□}(0) - S_{□}^{e}{}_{□□} \rVert_x$	0 (exact)
d3	$\lVert J^b(0) - B_{□}^{e}{}_{□□} \rVert_x$	0 (exact)
d4	$\lVert Ad(T) \cdot J^b - J_{□} \rVert_x$, arbitrary θ	1.82×10^{-15}
d5/d6	Both Jacobians are 6×6	confirmed

d7	$\|J^b - J\square^{uw}\square_e{}^{rerr}\|_x$ vs. finite differences	1.77×10^{-5}

At the home configuration, all preceding transforms are identity, so every column of $J\square$ should equal the corresponding screw axis—confirmed exactly (test d2). The adjoint relationship $J\square = Ad(T)J^b$ holds to machine precision (test d4). The finite-difference validation (test d7) perturbs each joint by $\varepsilon = 10^{-5}$ rad; the resulting error of 1.77×10^{-5} is consistent with the $O(\varepsilon)$ truncation error of a first-order finite-difference scheme.

Part (f) — singularity.m

Approach

Three analytical singularity conditions are checked, supplemented by an SVD catch-all:

- **Wrist:** $|\theta_5| < \varepsilon$ — joints 4 and 6 (both −X axes) become coaxial.
- **Shoulder:** wrist-centre XY-distance from the Z-axis < ε.
- **Elbow:** 2-link arm distance equals $L_1 + L_2$ (fully extended) or $|L_1 - L_2|$ (fully folded), where $L_1 = a_2 = 1.150$ m and $L_2 = \sqrt{(a_3^2 + \delta z^2)} \approx 1.001$ m.

The manipulability measure $w = \sqrt{(\det(J\square J\square^T))}$ is computed alongside for numerical confirmation.

Tests and Results

Test	Configuration	w	Detected	Type
f1–f2	$\theta_5 = 0$	2.68×10^{-9}	Yes	WRIST
f3–f4	$\theta_5 = \pi$	3.12×10^{-9}	Yes	WRIST
f6–f7	Non-singular [0.1,0.2,−0.3,0.4,0.5,0.6]	0.350	No	—
f8–f9	Arm fully extended, θ = [0,0,0,0,π/6,0]	5.89×10^{-2}	Yes	ELBOW
f10–f11	Wrist centre on Z-axis	9.54×10^{-4}	Yes	SHOULDER

Wrist singularities drive w to essentially zero (~10^{-9}), while a well-conditioned configuration yields w = 0.350. The elbow singularity at the fully-extended home pose is correctly identified via both the analytical condition ($d\square^W = L_1 + L_2 = 2.151$ m) and the SVD catch-all. The shoulder singularity test constructs θ_2 analytically to place the wrist centre on the Z-axis, accounting for the z-drop in the forearm.

Part (g) — Velocity Ellipsoids and Isotropy

Functions: ellipsoid_plot_angular.m, ellipsoid_plot_linear.m, J_isotropy.m, J_condition.m, J_ellipsoid_volume.m

Approach

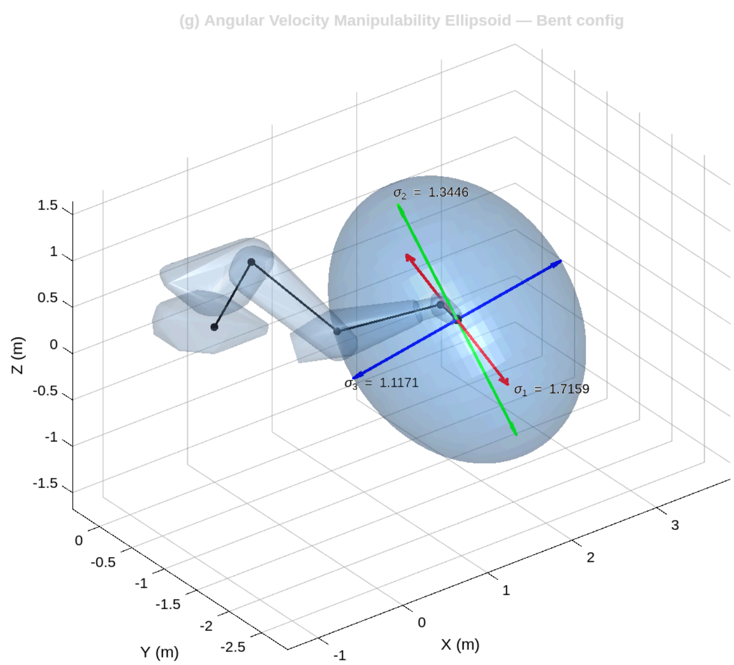
The space Jacobian is partitioned into angular ($J\omega$, rows 1–3) and linear (J_v, rows 4–6) sub-Jacobians. SVD yields semi-axes σ_i and principal directions U. The velocity ellipsoid is rendered as a parametric sphere scaled and rotated by U·diag(σ). The isotropy index is $\sigma\square_i\square/\sigma\square_a{}^x$; the condition number is $\sigma\square_a{}^x/\sigma\square_i\square$; and the ellipsoid volume is $(4/3)\pi\sigma_1\sigma_2\sigma_3$.

Tests and Results

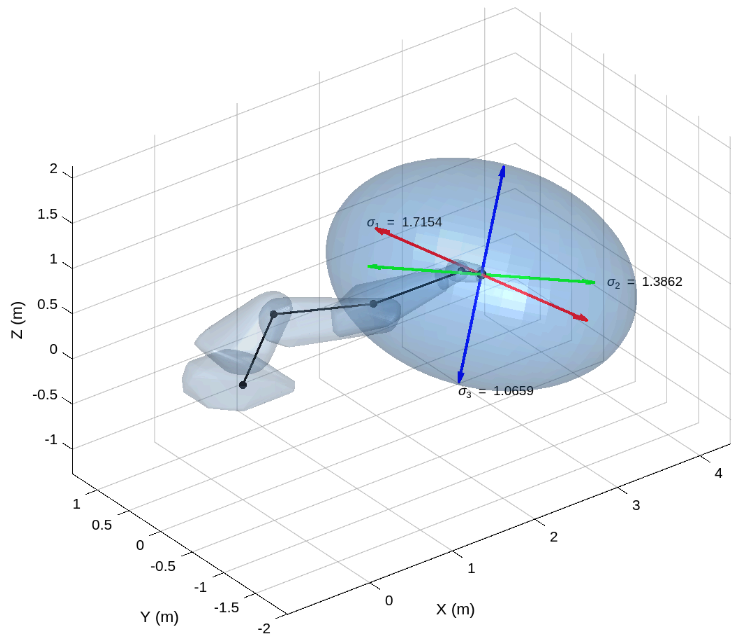
Evaluated at θ = [0.1, 0.2, −0.3, 0.4, 0.5, 0.6] rad (non-singular configuration):

Metric	Value
Angular semi-axes (σ_1, σ_2, σ_3)	1.7154, 1.3862, 1.0659
Angular ellipsoid volume	10.616
Linear semi-axes (σ_1, σ_2, σ_3)	2.9530, 1.5338, 0.5770
Linear ellipsoid volume	10.948
Isotropy index	0.02564
Condition number κ	38.999

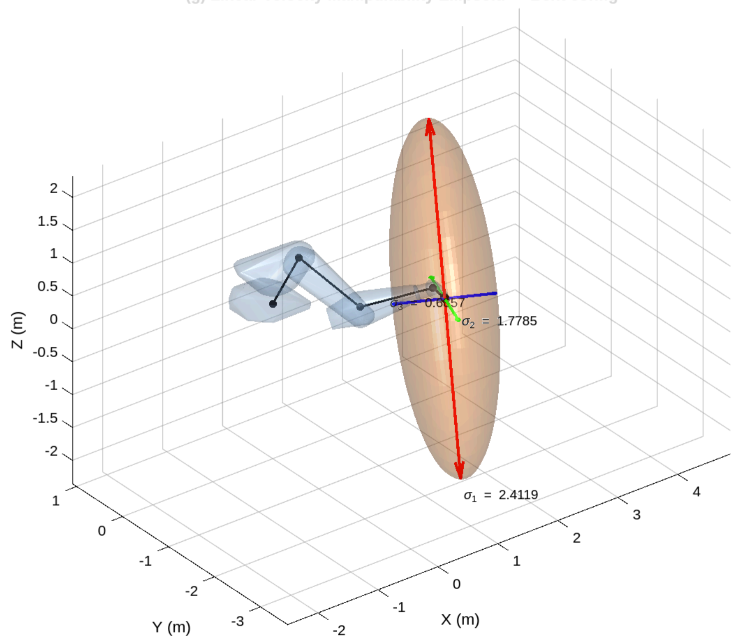
At a wrist singularity ($\theta_5 = 0$): isotropy → 2.65×10^{-17}, κ → ∞. Notably, the angular sub-Jacobian remains rank-3 at a wrist singularity (angular volume = 10.23 > 0), confirming that only the full 6-DOF Jacobian is rank-deficient there—the wrist retains full rotational freedom while losing one direction of independent linear motion.

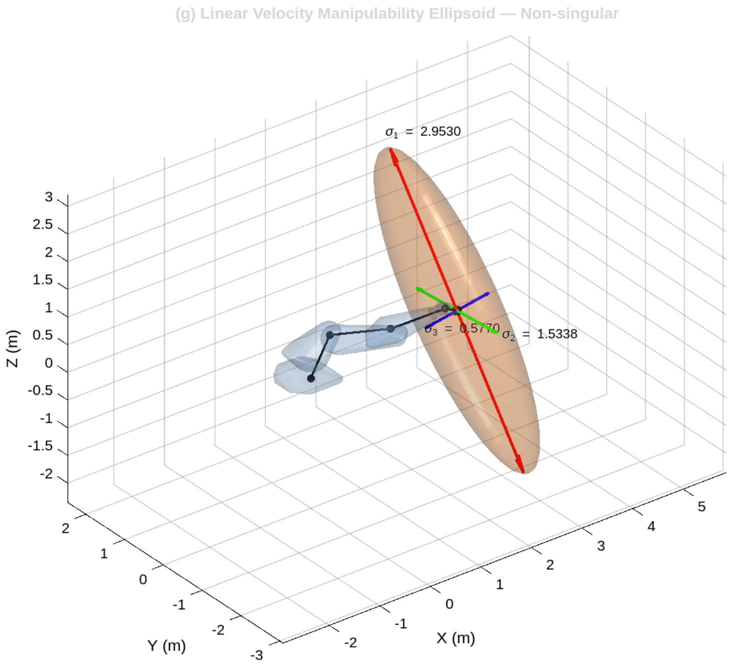


(g) Angular Velocity Manipulability Ellipsoid — Non-singular



(g) Linear Velocity Manipulability Ellipsoid — Bent config





Part (h) — J_inverse_kinematics.m

Approach

Newton–Raphson IK iterates the joint vector as:

$$\theta \leftarrow \theta + J^{b+}(\theta)\, V^b, \quad V^b = [\log(T^{curr}\, T^{d}{}_{e}\square)]^{v}$$

until $|\omega_e^{rr}| < \varepsilon\omega = 10^{-3}$ rad and $|v_e^{rr}| < \varepsilon_v = 10^{-3}$ m, or a maximum of 100 iterations is reached.

Tests and Results

Target: $\theta\square^{ru}{}_e$ = [0.1, 0.2, −0.3, 0.4, 0.5, 0.6] rad; initial guess: $\theta_0 = 0$

Test	Result
Convergence	Yes, 5 iterations
Angular residual $\lvert\omega_e^{rr}\rvert$	4.29×10^{-4} rad
Linear residual $\lvert v_e^{rr}\rvert$	4.91×10^{-4} m
$\lvert R\square^{o}\square - R^{d}{}_{e}\square\rvert_x$	6.07×10^{-4}
$\lvert p\square^{o}\square - p^{d}{}_{e}\square\rvert$	4.91×10^{-4} m
Unreachable target (10 m offset)	Correctly returns failure

Convergence in 5 iterations from a zero initial guess demonstrates the quadratic convergence characteristic of Newton–Raphson near the solution. Both angular and linear residuals are well below the 10^{-3} tolerance. The function correctly returns a failure flag when presented with an unreachable target.

Part (i) — J_transpose_kinematics.m

Approach

Jacobian-transpose IK replaces the pseudoinverse with the transpose:

$$\theta \leftarrow \theta + \alpha\, J^{bT}(\theta)\; V^b$$

with step size α = 0.1. This approach is computationally cheaper per iteration but converges more slowly than Newton–Raphson.

Tests and Results

Same target; initial guess: θ_0 = θ□ru$_e$ + 0.1

Test	Result
Convergence	Yes, 53 iterations
Angular residual	9.77×10^{-4} rad
Linear residual	5.78×10^{-4} m
Position error	5.78×10^{-4} m
Iterations vs. Newton–Raphson	53 vs. 5 (10× slower)
Unreachable target	Correctly returns failure

The Jacobian-transpose method converges to within tolerance but requires 53 iterations compared to 5 for Newton–Raphson, illustrating the expected trade-off: lower per-iteration cost at the expense of a substantially reduced convergence rate. A close initial guess was required owing to the narrower basin of attraction of the transpose method.

Part (j) — redundancy_resolution.m

Approach

Extends Newton–Raphson with a null-space secondary objective that maximises manipulability w = √(det(J□J□^{T})):

$$\theta \leftarrow \theta + J^{b+}\, V^b + (I - J^{b+} J^b)(k_0\; \nabla w)$$

For a 6-DOF robot, the null-space projector $N = I - J^{b+} J^b \approx 0$ and the secondary term vanishes, reducing to standard Newton–Raphson.

Tests and Results

Same target; $\theta_0 = 0$, $k_0 = 5$

Test	Result
Convergence	Yes, 5 iterations
Angular residual	4.29×10^{-4} rad

Linear residual	4.91×10^{-4} m
$\lvert N\rvert_x = \lvert I - J^{b+}J^{b}\rvert_x$	4.77×10^{-15} (numerically zero)
$\lvert T^{RR} - T^{□R}\rvert_x$	8.70×10^{-8} (solutions identical)
Unreachable target	Correctly returns failure

The null-space projector norm of 4.77×10^{-15} confirms that the 6-DOF robot has no kinematic redundancy—the secondary manipulability-gradient term contributes nothing, and the result matches the Newton–Raphson solution to $\sim10^{-8}$. The function is fully effective for redundant robots ($n > 6$) where the null space is non-trivial.

Part (k, Bonus) — DLS_inverse_kinematics.m

Approach

Damped Least Squares (Nakamura & Hanafusa, 1986) replaces the pseudoinverse with:

$$J^{DL□} = J^{bT}(J^{b}J^{bT} + \lambda^{2}I)^{-1}$$

Variable damping is applied: $\lambda = 0$ when $\sigma_{□i□} \geq \sigma_{□^{hr}e□^{h}}$ (reducing to standard Newton–Raphson), and $\lambda^2 = \lambda_{□a}^{x2}\,[1 - (\sigma_{□i□}/\sigma_{□^{hr}e□^{h}})^2]$ near singularities.

Tests and Results

Test	Configuration	Result
k1–k4	Non-singular target $\theta_{□^{ru}e}$	Converged in 4 iterations; $\lvert p_e^{rr}\rvert = 8.20\times10^{-5}$ m
k5	Well-conditioned ($\lambda \approx 0$): compare to NR	$\lvert T^{DL□} - T^{□R}\rvert_x = 1.04\times10^{-3}$ (within IK tolerance)
k6	Near-wrist singularity ($\theta_5 = 0.01$ rad)	Converged in 13 iterations
k7	Unreachable target	Correctly returns failure

On a well-conditioned target, DLS matches Newton–Raphson (4 vs. 5 iterations, solutions within mutual IK tolerance). Near the wrist singularity ($\theta_5 = 0.01$ rad, $\sigma_{□i□} \ll \sigma_{□^{hr}e□^{h}}$), damping activates and the algorithm still converges in 13 iterations—a configuration where standard Newton–Raphson would amplify errors and diverge. This confirms the core benefit of DLS: graceful degradation near singularities at the cost of a modest reduction in accuracy.

Part (m, Bonus) — Graphical Simulation

Functions: robot_simulation.m, robot_animation.m

Two graphical simulations were implemented. `robot_simulation.m` animates the KR120 tracing a tilted ellipse (centre [1.7, 0, 1.0] m, semi-axes 0.5×0.5×0.35 m) using DLS IK at 12 Cartesian waypoints, interpolated with cubic joint-space splines at 30 frames per waypoint. The manipulability measure w is computed at each frame and displayed live via a colour-coded annotation.

`robot_animation.m` demonstrates four end-effector paths—circle, square, helix, and right-angle snake—all at depth $x = 1.9$ m, using DLS IK with a 300-iteration budget and $\lambda_{□a}^{x} = 0.15$, $\sigma_{□^{hr}e□^{h}} = 0.05$. Each animation overlays the translucent STL mesh, the full desired path, the traced end-effector path, joint coordinate frames, and a real-time end-effector marker. Animations are exported as Motion JPEG AVI files.

Both simulations use `stlread` to load the seven collision-mesh STL files from the `kuka_experimental` ROS package, and align them to the Product of Exponentials skeleton via pre-computed home-configuration offsets derived directly from the URDF joint origins.

Video files can be found in the associated github repository for review.

AI Disclosure)

For this assignment, Anthropic's Claude was used for assistance. Validation was done via the test cases, review of the code written by Claude, and graphical representations that could be reviewed visually.

Included is a collection of matlab functions, urdf files, stl files, and a pdf. The pdf describes a programming problem starting at page 3. Please present a plan for addressing assignment steps (a) through (k) along with the relevant test functions. Do not use the stl files for now when producing graphics. Go step by step (a, b, c, etc). First propose a course of action for each step, and wait for approval before starting to generate code. Once approved, stop once that step is done and wait to move onto the next. When possible, use the matlab functions already provided.

Please write helper functions to generate all images and plots in one command and to run all tests in one command.

The graphics output does not seem correct. Please re-review the provided reference files to ensure that the proper robot specifications are used.

Now, please incorporate the stl files provided into the graphics rendering.

Please create 4x animations of the robot tracing a circle, a square, a helix, and a square snake pattern. The animations should be able to be saved as video files. The animations should have a dotted line where the robot should trace, and a solid line where the robot traces. All simulation should be done using the functions already written for this project.

There is an error with the animation program as it tries to save the files.

Please help provide a basic framework and rough draft for a discussion of each point a-m in the attached pdf file. There should be a section for the approach for the section and any test cases for the functions named in the pdf.

Please refactor this attached file so that the scientific/latex symbols and notation are done properly and so that the file is tidied up.